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Q1. Multiple Choice Questions (5)

1. **What geometric configuration of two lines in a 2-variable system of linear equations corresponds to a unique solution?**
   a) The lines form a right angle.
   b) The lines are coincident.
   c) The lines intersect at exactly one point.
   d) The lines are parallel and never intersect.
2. **In the Gaussian Elimination process, the phase where all entries below the main diagonal of the augmented matrix are made zero, column by column, is called:**
   a) Partial pivoting
   b) LU factorization
   c) Backward substitution
   d) Forward elimination
3. **If you need to solve Ax=b for many different right-hand side vectors b but the same matrix A, why is LU factorization efficient?**
   a) Forward and back substitution for each new b is $O(n^3)$, which is the same cost as LU factorization.
   b) The $O(n^3)$ LU factorization is computed once, and each subsequent solution using forward/back substitution is only $O(n^2)$.
   c) LU factorization is only used for symmetric matrices, which automatically simplifies the solution process.
   d) The LU factorization cost is $O(n^2)$, which is faster than Gaussian Elimination's $O(n^3)$.
4. **What is the main numerical issue that is solved by applying pivoting in the LU factorization process?**
   a) It reduces the overall computational complexity of the LU factorization from $O(n^3)$ to $O(n^2)$.
   b) It swaps rows to avoid division by a zero or very small pivot element, thus preventing rounding errors from exploding.

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c) It makes the matrix diagonally dominant, guaranteeing convergence for iterative methods.
d) It ensures that the resulting L and U matrices are symmetric.

5. **What is the procedure in Scaled Partial Pivoting to determine which row to choose as the pivot row at step k?**
   a) Pick the row with the largest absolute value in the pivot column.
   b) Pick the row with the smallest difference between the pivot column entry and the largest number in that row.
   c) Pick the row with the largest absolute value in the entire remaining submatrix.
   d) Compare the number in the pivot column divided by the biggest number of that row and pick the row with the biggest scaled value.

Q2: True/False **(5)**

1. The goal of Gaussian Elimination is to convert the augmented matrix into a lower triangular form, followed by backward substitution. (T / F)
2. Partial pivoting in LU factorization involves swapping both rows and columns to find the largest pivot element. (T / F)
3. A symmetric matrix is guaranteed to be positive definite if it is also diagonally dominant and has positive diagonal entries. (T / F)
4. In the Gauss-Seidel iteration, when calculating the next approximation for a variable x_i, the most recently computed values for x_j (where $j<i$) are used. (T / F)
5. For the Successive Over Relaxation (SOR) method, using a relaxation factor omega such that $0 < \text{omega} < 1$ is known as over-relaxation and results in faster convergence. (T / F)

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Q3: Short Answer Questions (5)

1. **Briefly explain the two main phases of the Gaussian Elimination process and state the objective of each phase.**

2. **Explain the difference in the procedure for Partial Pivoting versus Full Pivoting**

3. **State two conditions on matrix A that, if satisfied, guarantee the convergence of the Gauss-Seidel iterative method for any initial vector x^0.**

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4. **In the context of LU factorization without pivoting, explain the numerical issue that arises when a pivot a_{ii} is a very small number, even if it is non-zero.**

5. **What is the range of the relaxation factor omega in the Successive Over Relaxation method that is known as over-relaxation, and what is the typical value used for optimal performance?**